

A spatial machine-learning model for predicting crop water stress index for precision irrigation of vineyards

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ABSTRACT

Optimization of water inputs is possible through precision irrigation based on prescription maps. The crop water stress index (CWSI) is an indicator of spatial and dynamic changes in plant water status that can serve irrigation management decision-making. The driving hypothesis was that in-season CWSI maps based on combined static and spatial-dynamic variables could be used to delineate irrigation MZs. A primary incentive was to minimize thermal-imaging campaigns and to complement CWSI maps between campaigns with cost-effective multi-spectral imaging campaigns producing normalized difference vegetative index (NDVI) maps. A spatial machine-learning model based on a random-forest (RF) algorithm combined with spatial statistical methods was developed to predict the spatial and temporal variability in CWSI of single vines in a vineyard. Model criteria and objectives included the reduction of sample data and input variables to a minimum without impacting prediction accuracy, consideration of only variables readily available to farmers, and accounting for spatial location and spatial processes.

The model was developed and tested on data from a 'Cabernet Sauvignon' vineyard in Israel over two years. Prediction of CWSI was driven by terrain parameters, slope, aspect and topographical wetness index, soil apparent electrical conductivity (ECa), and NDVI.

Spatial models based on RF were found to support CWSI prediction. Adding a geospatial component significantly improved model performance and accuracy, particularly when raw data was represented as z-scores or when z-scores were used as weights. NDVI, followed by ECa, aspect, or slope, was the most important variable predicting CWSI in the non-spatial models. The stronger the variable importance of NDVI, the better the model performed. The weaker the effect of NDVI in predicting CWSI, the stronger the effect of terrain and soil variables. In the spatial models, based on z-transformed values or on weighted values, the most important variable in predicting CWSI was either NDVI or location.

The model, based on a limited and readily accessible number of variables, can serve as the basis for user-friendly decision support tools for precision irrigation. Additional research is needed to evaluate alternative prediction variables and to account for case studies in more geographical locations to address overfitting specific input data. Socio-economic and cost-benefit considerations should be integrated to examine whether precision irrigation management based on such models has the desired effects on water consumption and yield.

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1. Introduction

Environmental considerations and depletion of water resources are major driving forces in efforts to optimize water use in agriculture, particularly in arid and semi-arid regions. For these reasons, the development of precision agriculture (PA) and precision irrigation (PI) methods and practices has been accelerated. An important aspect of PA is location- and crop-based management of input resources. This can be achieved by quantifying within-field spatial variability and in- and between-season temporal variability as a basis for dividing a field into management zones (MZs) on which spatially and temporally variable rate application of inputs is applied (González Perea et al., 2018). Precision agriculture has been effective in reducing and optimizing resources such as fertilizers, energy, or water (González Perea et al., 2018; O'Shaughnessy et al., 2016; Ohana-Levi et al., 2019a; Stone et al., 2020). Optimization of water application at a MZ scale has been shown to positively affect the quality and quantity of crops while increasing irrigation water productivity (Batchelor et al., 2002; Stone et al., 2020).

Existing models for optimizing irrigation inputs often focus on approaches based on delineating MZs that are static over time, such as methods developed by Kitchen et al. (2005), Córdoba et al. (2013), Peeters et al. (2015), Ohana-Levi, et al. (2019a) and Bahat et al. (2021). Management of precision irrigation and irrigation MZs can benefit from accounting for variability and in-field dynamic changes over time in crop water status both in-season and between-seasons (Cohen et al., 2017a; O'Shaughnessy et al., 2015; Fontanet et al., 2020). In order to achieve spatial-temporal optimization of water inputs, it is important to detect indicators (i.e., response variables) that can support the determination of an optimal amount of irrigation.

Assessment of plant water status across the field is often a prerequisite for precision irrigation (Bellvert et al., 2014). Leaf water potential (LWP, Ψ_{leaf}) or stem water potential (SWP, Ψ_{stem}) are widely used crop indicators of plant water status or stress. Current best management practices (BMPs) for irrigation management and decisions in vineyards (Bahat et al., 2021; Poblete-Echeverría et al., 2017), in other intensively cultivated tree crops (Shackel et al., 2010) and in some field/vegetable crops (Cohen et al., 2005) are based on LWP or SWP measurements and thresholds dictated by the phenological state of the crop (Bahat et al., 2021; Zinkernagel et al., 2020). While LWP is determined on single, uncovered leaves, SWP is measured on a pre-covered leaf or set of leaves, in equilibrium with and therefore representing the entire plant. SWP is most commonly used to quantify water status in trees and vines (Munitz et al., 2018; 2020) and will be referred to exclusively in this paper.

In-situ SWP measurements, typically collected manually using a pressure bomb (Bahat et al., 2021; Boyer, 1995), are destructive, time-consuming, and labour intensive. Consequently, often very few point measurements are collected in a given field or during a growing season, limiting their spatial and temporal representation. SWP data are used for decision-making but, due to the limited sampling, spatial variability cannot be captured, and quantification of dynamic changes across time is limited. This suggests that an alternative approach should be adopted in which an indicator that can be readily collected with high spatial and temporal resolution should be used in order to represent spatial variability in plant water status. Plant condition indices, derived from remotely-sensed collected data, such as the normalized difference vegetation index (NDVI) and the crop water stress index (CWSI), acquired using unmanned aerial vehicles (UAVs), offer both high-spatial and -temporal resolution (Santesteban et al., 2017).

CWSI, based on canopy temperature computed from thermal imagery, has been used since the 1980 s for measuring crop water status and the effect of irrigation on its spatial variability (Jackson et al., 1981). CWSI has been found to be a good indicator of plant water status and plant stress conditions (Cohen et al., 2017c; Khanal et al., 2017; Lacerda et al., 2022; Matese et al., 2018; Meron et al., 2010; Poblete-Echeverría et al., 2017; Rud et al., 2014) and to be correlated with SWP values in vines and in a number of other fruit tree crops (Bellvert et al., 2014;

Möller et al., 2007; Katz et al., 2023). CWSI has become an acceptable index to map within-field variability of crop water status using aerial thermal images (e.g., Gonzalez-Dugo et al. (2013), Cohen et al. (2017b), Bahat et al. (2021), and Katz et al. (2023)). As thermal imagery is collected over a short period of time with potential for high temporal resolution during the growing season, CWSI can be calculated for each pixel in an image offering spatial-temporal dynamic representation of crop water deficit stress. This suggests that CWSI can support irrigation decisions or be used as a partial proxy for SWP field measurements, thus reducing the number of required SWP in-situ measurements during the season. CWSI maps can be used as a key input to produce dynamic irrigation MZ maps (Bahat et al., 2024). Nonetheless, while UAV multi-spectral imaging campaigns that deliver high resolution NDVI maps have become cost-effective and extensively used by farmers, UAV thermal imaging campaigns require substantial financial investments that currently inhibit their large-scale adoption (Klapp et al., 2021; Olson & Anderson, 2021).

Static predictors are those derived from soil survey maps or from topographical attributes (Corwin & Plant, 2005; Kitchen et al., 2005; Sudduth et al., 2010; Vitharana et al., 2008) and represent terrain and soil properties. Dynamic predictor variables are those derived from remote-sensing, representing crop status, such as NDVI and CWSI, based on aerial multi-spectral and thermal-imaging, respectively. Potentially, combining dynamic and static variables could represent in- and between-season fluctuations in plant water status and therefore produce real-time dynamic spatial-temporal precision irrigation maps (Cohen et al., 2017b; Matese et al., 2018; Stone et al., 2020).

The driving hypothesis of this research was that in-season CWSI maps based on combined static and spatial-dynamic variables could be used to delineate irrigation MZs. A main incentive was to minimize thermal-imaging campaigns and to complement CWSI maps between campaigns with NDVI maps, produced by the more cost-effective multi-spectral imaging campaigns. The primary objective therefore was to develop a model predicting CWSI. Secondary objectives included: 1) Reduction of input variables to a minimum without significantly impacting the accuracy of the prediction model; 2) Consideration of only variables readily available to farmers (not time-consuming or expensive) and; 3) Accounting for spatial location and spatial processes in the prediction model.

To understand the relations between static and dynamic variables and quantify their effect on the spatial and temporal variability of CWSI, methods from spatial statistics and machine-learning (ML) were employed. Complex systems such as those in agriculture tend to exhibit non-linear and non-stationary relations among variables. This suggests that an approach, which can quantify complex interrelations in prediction models based on statistical algorithms, is required. ML methods such as boosted regression trees (BRT) and random forests (RF), (Biau & Scornet, 2016; Breiman, 2001) are suitable for capturing complex, non-linear, relations in multivariate systems and have become popular for analyzing complex spatial phenomena, including agricultural data (Chlingaryan et al., 2018; Georganos et al., 2019; Hengl et al., 2018; Li, 2019; Ohana-Levi et al., 2019b). However, unlike spatial statistical approaches (Ohana-Levi et al., 2019a; Peeters et al., 2015), these methods do not consider the location of features in their algorithms. Due to the nature of agricultural processes, the adopted approach should additionally account for heterogeneity in spatial processes, including spatial autocorrelation (Vitharana et al., 2008), since such processes, if significant, might introduce bias to the predictions (Li, 2019). The challenge is, therefore, to introduce location-based processes into ML methods. A few attempts to do so have been made by combining, in the algorithms, the neighbouring values and the distances between features. Sekulić et al. (2020), for example, developed an RF spatial interpolation method by introducing covariates to the model and including n nearest observations and distances to those observations. Georganos et al. (2019) developed a geographical random forest (GRF) model by disaggregating RF into local sub-models. Fox et al. (2020) presented RF regression

kriging. Li (2019) combined a geographically weighted regression model with RF, and Talebi et al. (2022) developed a spatial RF algorithm by increasing the dimension of input data to represent spatial heterogeneity and spatial dependencies. Finally, optimization of precision irrigation practices, particularly prediction models based on ML methods, requires an approach based on analysing a relatively large number of input variables with sufficient spatial coverage, which are often not readily accessible to farmers.

To address these challenges, we focused on developing an applicable spatial prediction model to quantify the spatial variability of CWSI, support precision irrigation, and serve as a basis for future development of irrigation management decision support tools (DSTs). In addition, to allow the development of a method based on a limited number of readily accessible variables, a heuristic approach was adopted. Such an approach was projected to present a rational trade-off in which the input number of variables is reduced, thus lowering cost and increasing applicability without significantly compromising the accuracy of the prediction.

2. Materials and methods

The research focused on a model developed to quantify the spatial variability of water stress in a wine-grape vineyard in the area of the Judean Hills near Jerusalem, Israel, which was host to a precision irrigation field trial. First, the standard BMP used for characterizing water stress and for irrigation decisions in wine-grape vineyards is presented. This BMP follows the practice in which data from midday SWP measurements indicate water stress and guide irrigation management decisions. This is followed by establishing, based on previous research, remote-thermal image-based CWSI as a proxy for SWP. Last, a CWSI prediction model is proposed that outputs maps representing spatial changes in water status across the vineyard. The developed model integrates advanced ML methods with methods for multivariate spatial-statistical analysis. Spatial considerations introduced to the CWSI prediction model give rise to a spatial machine-learning (SML) model, which can provide CWSI predictions at the single vine and MZ scales and support precision irrigation.

2.1. Case study

The experimental vineyard is situated in the Judean Hills 10 km from Jerusalem, next to the village Mevo Beitar, at coordinates 31°43' N; 35°06' E (Fig. 1). The area is characterized by a Mediterranean climate, with a mean maximum summer temperature of 29.4 °C, a mean minimum winter temperature of 6.4 °C and a mean annual precipitation of 540 mm occurring exclusively between October and May. The area's lithology is composed of limestone, mixed with chalk and marl, and covered with shallow terra-rosa soil. Topography is dominated by sloping hills at elevation ranging between 675–900 m above sea level. The area of the vineyard is characterized by high variability of slopes and aspects (orientations). A distinct gradient exists from east to west with higher elevation in the east. In addition, north and south-facing slopes form a wadi that crosses the vineyard from east to west (Fig. 1). The prevalent surrounding vegetation is a mixture of natural Mediterranean low shrubs and trees and planted pine trees. The vineyard comprises 5522 *Vitis vinifera* L. 'Cabernet Sauvignon' vines grafted onto 101–14 rootstock on a 2.5 ha area. Spacing in the vineyard is 1.5 m between vines and 3 m between rows. The vines were planted in 2011 with row direction of 117–297. Rainfall prior to the growing seasons reached 369 mm in the winter of 2016–2017 and 452 mm in 2017–2018. During the 2017 growing season, the vineyard was drip-irrigated uniformly once a week until harvest with a seasonal total of 98 mm. During the 2018 growing season, the vineyard was managed with variable rate irrigation (VRI) based on 20 equally sized (30 x 30 m) management cells (Fig. 1). The eastern part of the vineyard (MCs N–T) was irrigated uniformly with a total seasonal application of 32 mm, while the western part of the vineyard received VRI ranging from 23 to 66.2 mm. The 2.3 ha area containing all 20 MCs with a total of 3893 vines was used for the model development.

2.2. Data collection and preprocessing

Variables included in the prediction model were chosen based on the objective of selecting only those most readily accessible to growers (not time-consuming or costly). Chosen variables include both those considered constant with little change over time (static variables) and those considered dynamic variables, potentially changing in time

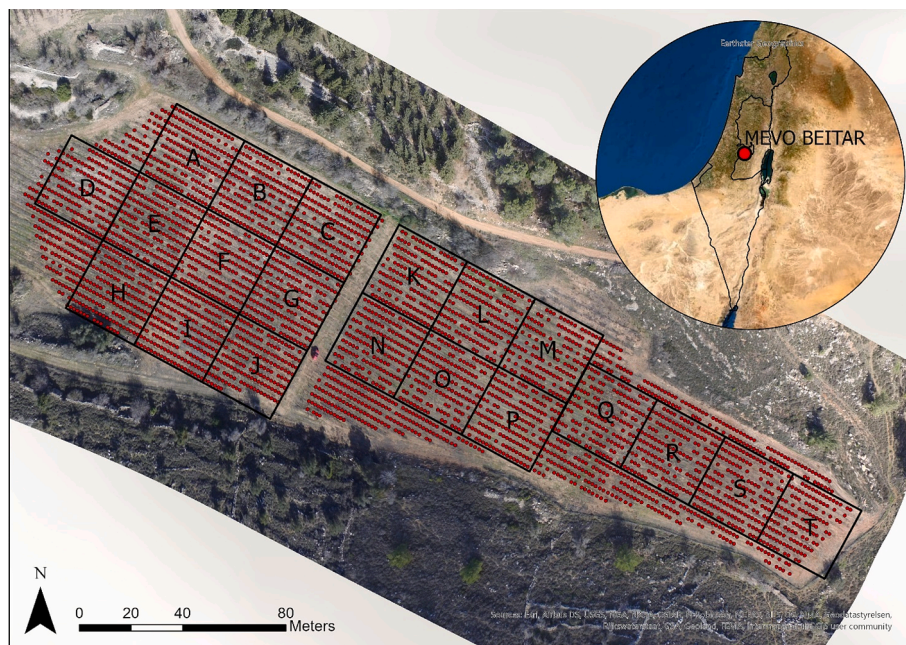


Fig. 1. Study site of the Mevo Beitar vineyard. General location of vineyard in Israel (right circle). Study site layout with red dots representing single vines and black rectangles indicating management cells for precision irrigation, overlaid on an orthophoto with 0.125 m spatial resolution (image acquired from UAV in 2017).

(Ohana-Levi et al., 2021). Input static variables included (Fig. 2):

Terrain:

- Slope and aspect, which have been found to affect crop water availability (Geroy et al., 2011). Aspect (orientation of slope in degrees from North) and slope (in degrees) were derived from a digital elevation model (DEM) with a 1 m spatial resolution and

were calculated for each pixel in the DEM. Topographic wetness index (TWI), which is a function of both the downslope water drainage represented by the slope gradient and the upstream catchment contributing area and is often used to represent the distribution of soil moisture conditions (Bahat et al., 2021; Beven & Kirkby, 1979; Raduła et al., 2018). TWI was computed from a DEM with 1 m spatial resolution and

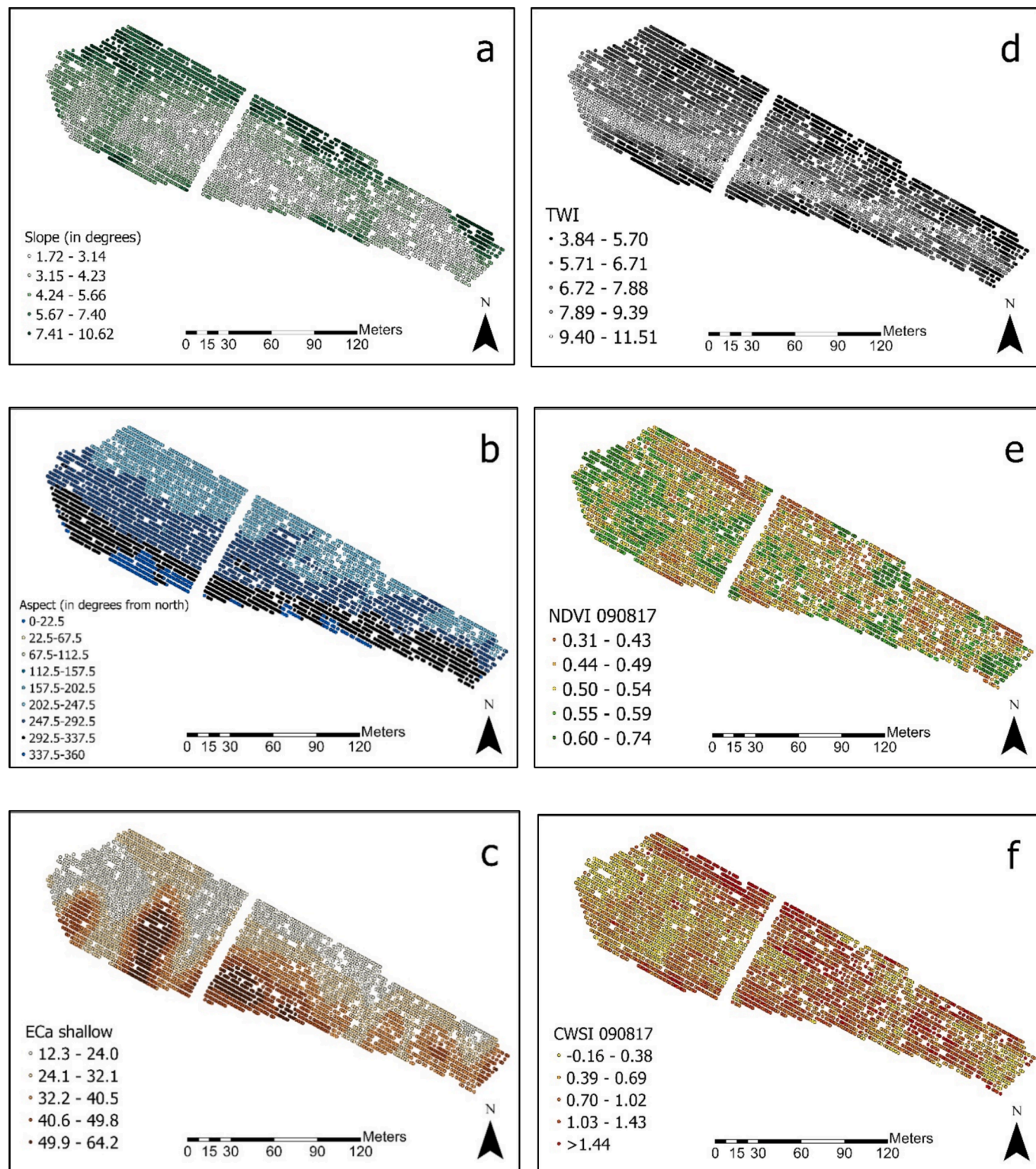


Fig. 2. Static and dynamic input variables extracted for each vine location: (a) slope, (b) aspect, (c) ECa, (d) TWI, (e) NDVI and (f) CWSI. NDVI and CWSI values are from August 9, 2017.

calculated for each pixel using the following equation (Sørensen et al., 2006): $TWI = \ln(A/\tan\beta)$, where A is the specific area, or the upstream catchment contributing area (m^2) for each pixel, and β is the local slope gradient in the steepest downslope direction in degrees.

Soil:

- Apparent electrical conductivity (ECa) [mS/m]. A measure of a soil's ability to conduct electricity, ECa values indicate soil physical and chemical properties, including texture/clay content, moisture, salinity, cation exchange capacity, bulk density and organic matter content. ECa is relatively easy to collect and has been shown to be a reliable indicator of soil productivity and soil spatial variability. ECa is often used as a basis for delineating MZs (Corwin & Plant, 2005; Kitchen et al., 2005). ECa was collected using a four-probe soil resistance sensor (Soil EC 3100, Veris Technologies). Data recorded from the shallow layer (0–0.30 cm) were used to derive an interpolated ECa surface map.

Input dynamic variable, NDVI (Fig. 2):

- NDVI was calculated from the red and near-infrared bands of a multi-spectral image with a spatial resolution of 7–8 cm. NDVI has been used extensively as an indicator for crop vigor and yield (Quarmby et al., 1993). In addition, research has suggested correlations between NDVI and SWP in Mediterranean vineyards as well as correlations between NDVI and plant water status and CWSI, although such correlations are not always consistent throughout the season (Gonzalez-Dugo et al., 2013; Helman et al., 2018). Remotely sensed images were collected with a multi-spectral camera MicaSense RedEdge MX (MicaSense® Inc, Seattle, USA) mounted on a UAV (Datamap Group, Bnei Brak, Israel) with a flight height of 100 m. Scanning campaigns were conducted on August 9, 2017; September 5, 2017 (five days prior to harvest); July 4, 2018; August 15, 2018, and August 29, 2018 (six days prior to harvest) midday, between 12:00 and 13:00 Israeli Standard Time. NDVI was derived based on Eq. (1) (Rouse et al., 1974; Tucker, 1979):

$$NDVI = \frac{\rho(NIR) - \rho(red)}{\rho(NIR) + \rho(red)} \quad (1)$$

- where $\rho(NIR)$ is the spectral reflectance value for the near-infrared band, and $\rho(RED)$ is the spectral reflectance value for the red band. NDVI values range from –1 to 1. To assign the NDVI values to each specific vine and make sure that mainly pixels belonging to the canopy were accounted for and not background soil pixels, a buffer with a radius of 20 cm was defined around each vine point. This ensured that 24 neighboring pixels primarily representing the canopy were taken into account. The value associated with each vine was calculated as the mean NDVI value within the buffer.

Predicted variable CWSI (Fig. 2):

- CWSI was calculated from a thermal image with a spatial resolution of 7 cm. Images were collected with a UAV on the same dates and time as the NDVI images using a FLIR SC2000 thermal camera (FLIR® Systems, Inc., Billerica, MA, USA). To compute CWSI, collected images were first mosaicked, the soil was removed using a histogram bimodal segmentation of the temperature, and the shaded area adjacent to the vines was masked out using the NDVI image in order to extract pure canopy pixels (Bahat et al., 2021). Computation of CWSI was

based on the normalization of canopy temperature values (Jackson et al., 1981; Jones, 1992) – Eq. (2):

$$CWSI = \frac{T_c - T_w}{T_d - T_w} \quad (2)$$

- where T_c is the actual canopy temperature (for each pixel), derived from the thermal image. T_w is the temperature of a fully-transpiring canopy which was calculated by averaging the lower 5 % of the temperature values in the thermal image (Cohen et al., 2017c) and T_d is the temperature of a non-transpiring canopy which was computed as T_a (air temperature) + 7C. T_a was derived from an in-field meteorological station (Meteotech, Israel). To assign the temperature in the thermal image to a specific vine canopy and validate that mainly pixels that belong to the canopy were accounted for and not soil or other background pixels, a 20 cm buffer was defined around each vine point, as per the NDVI calculations. This ensured that 24 neighboring pixels primarily representing the canopy were considered in the computation of the CWSI. Mean CWSI values ranged between 0.52 on July 4, 2018 to a maximum of 0.95 on August 29, 2018. It is well-accepted in the literature that CWSI values in vines can vary greatly between 0.2 and less for well-watered vines to 0.7 and even as high as 1.0 and 1.20 for severely stressed vines (Bellvert et al., 2014; Matese et al., 2018).

2.3. Model development

An RF-based model predicting CWSI as a response variable for the entire vineyard was developed. The model was tested for its performance to evaluate variations in accuracy during the growing season. To account for location and address geospatial processes, three different geospatial approaches were introduced to the RF model and used to develop the SML model:

- a. Representation of spatial location by introducing the (x, y) coordinates [m] as predicting variables using a Cartesian transformation (Israel TM projection – ITM). These variables were derived for each vine location. In previous research, adding (x, y) coordinates was found to improve ML model performance by accounting, in part, for spatial autocorrelation, especially when training data well represents the geographical space (Li, 2019). To evaluate the combined effect of the (x, y) coordinates and characterize the interaction of both coordinates, their product (y/x) was used as one variable as proposed by Li (2019).
- b. Representation of spatial autocorrelation by assigning to each point a local measure using the Getis Ord Gi* statistic (Getis & Ord, 1992; Ord & Getis, 1995) transformed into z-scores (Ohana-Levi et al., 2019a; Peeters et al., 2015). A detailed explanation of the Getis Ord Gi* statistic is given in Appendix A. This enables accounting for the spatial structure by representing the relation of each feature value to its neighboring values as a function of distance. Each variable at each point location is transformed into z-scores.
- c. Representation of spatial autocorrelation using the z-scores calculated in the previous approach as spatial weights and weighing each point value (raw data) by its associated z-score.

The RF model was developed using Leo Breiman's ML RF algorithm (Eq. (3)), a supervised ML method based on bagging, in which the training dataset is split into smaller groups of data or decision trees (Biau & Scornet, 2016; Breiman, 2001; Breiman et al., 1984; Fox et al., 2020; Hengl et al., 2018). The trees together, or the entire forest, are used to create the final predictions instead of predictions based only on any single tree. Prediction is based on averaging across the decision trees:

$$\hat{f}(x) = \frac{1}{B} \cdot \sum_{b=1}^B f_b(x) \quad (3)$$

Where b is the individual bootstrap sample; B is the total number of trees and f_b is the individual decision tree.

RF models have several advantages for coping with complex datasets: they are indifferent to non-linear relations between variables, missing data, outliers, and multicollinearity; and they can address a variety of predictor variables such as binary, categorical, and numerical variables and they do not require a-priori rescaling or transformation of data (Biau & Scornet, 2016; Chlingaryan et al., 2018). The fact that the entire forest is considered for predictions and that, in each split, a random subset of the training data and of the explanatory variables is used, corrects the tendency of decision trees to over-fit their training dataset. In addition, RF models are often used for modelling spatial data since they are completely non-parametric and are indifferent to spatial autocorrelation (Hengl et al., 2018). The spatial RF model was developed using Python (Van Rossum and Drake, 2009) and ArcGIS Pro (ESRI, 2019) and includes the steps outlined in Fig. 3:

Data preprocessing and data extraction of static and dynamic attribute values associated with each single vine following the protocol described in section 2.2.

Adding the (x, y) coordinates data as an additional predicting variable (y/x).

Applying the Getis-Ord G_i^* statistic to each variable (prediction and response variables, excluding the (x, y) variables) and transforming them to z-scores (Ord & Getis, 1995) for each data point (x_i) – termed $G_i^*(x_i)$.

Applying the G_i^* at point (x_i) as spatial weights to each variable (prediction and response variables, excluding the (x, y) variables), by multiplying the $G_i^*(x_i)$ times the raw data value at each data point (x_i) – termed $(G_i^*(x_i)) \cdot (x_i)$.

Training the Spatial RF model (fitting the model to the training data), evaluating its performance, and selecting the best-performing model. This includes randomly splitting the data into two subsets, 75 % of the data is used for the Spatial RF training and 25 % as a testing dataset for validation. The model is trained without the validation data. Model training was done for each of the first three Spatial RF geospatial approaches a-c: accounting for location by adding (x, y); accounting for spatial autocorrelation by representing values as z-scores and accounting for spatial autocorrelation by weighing values by their associated z-scores. The models and their performance were

evaluated in several ways: (i) by making predictions on the validation data and comparing the predictions with the known values from the validation dataset, (ii) by using the out-of-bag (OOB) errors (Breiman, 2001) by which each tree in the model received only a random subset of the training data, the excluded data was then run through the tree to examine how well the tree could predict the data, and (iii) by inspecting the percentage of variable importance in order to understand which variables are most important in the prediction. As a reference for evaluating performance, models were also trained on the raw variables, without any spatial information. The following accuracy measures were used: R^2 , which measures how well the model predicts, and mean absolute error (MAE), mean squared error (MSE) and root mean squared error (RMSE), which are metrics commonly used for evaluating regression models. The best-performing model was selected and fine-tuned by adjusting its hyperparameters.

Predicting CWSI for a different date: the best performing model was applied to a different date to predict CWSI values for each vine.

Decision on irrigation based on predictions and CWSI thresholds.

It is important to note that predictions based on the $G_i^*(x_i)$ values for each variable actually represent the predicted z-transformed (G_i^*) values of CWSI. Similarly, predictions based on the weighted $(G_i^*(x_i)) \cdot (x_i)$ values for each point represent the predicted $(G_i^*(x_i)) \cdot (x_i)$ values of CWSI. To derive the predicted raw CWSI value at each point (x_i), the following calculation can be applied to the predicted values (Eq.4):

$$CWSI_i = \frac{(G_i^*(x_i)) \cdot (x_i)}{G_i^*(x_i)} \quad (4)$$

Where ($CWSI_i$) represents the predicted CWSI value at point i ; $G_i^*(x_i)$ represents the predicted z-transformed value at point i and $(G_i^*(x_i)) \cdot (x_i)$ represents the CWSI value at the i^{th} point weighted by the z-transformed value at the i^{th} point. A more detailed explanation regarding back-transforming the z-transformed values into raw CWSI values is found in Appendix B.

3. Results and discussion

Relationships between all variables were first examined using a correlation matrix. For data from August 9, 2017 (Fig. 4) the strongest correlations were found between NDVI and CWSI (0.81), between slope and TWI (0.62) and between slope and ECa (0.56). This pattern was also observed in the correlation matrices of the data from September 5, 2017,

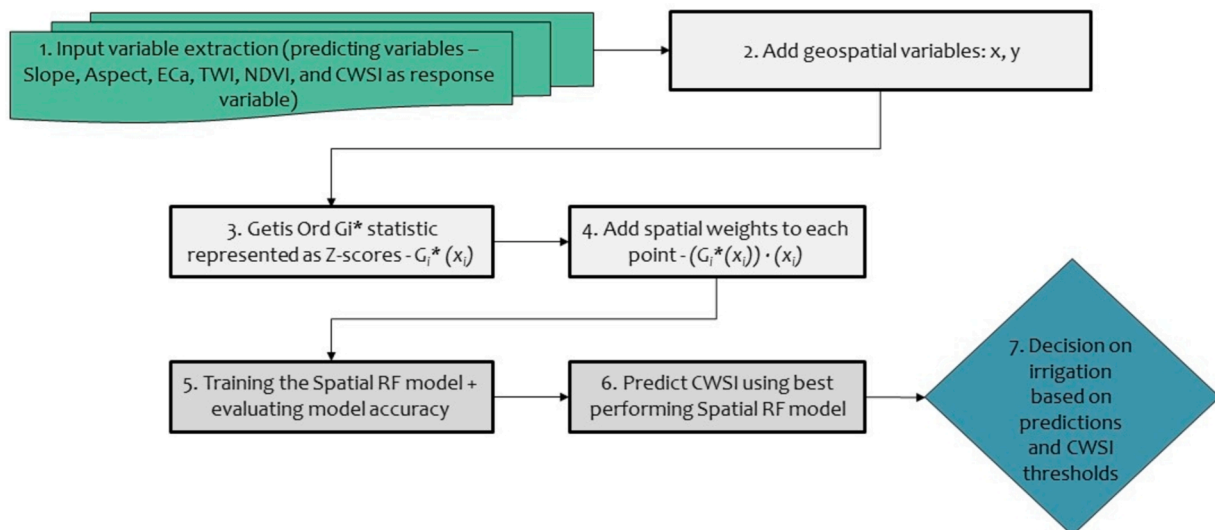


Fig. 3. Schematic flow chart of the spatial random forest model.

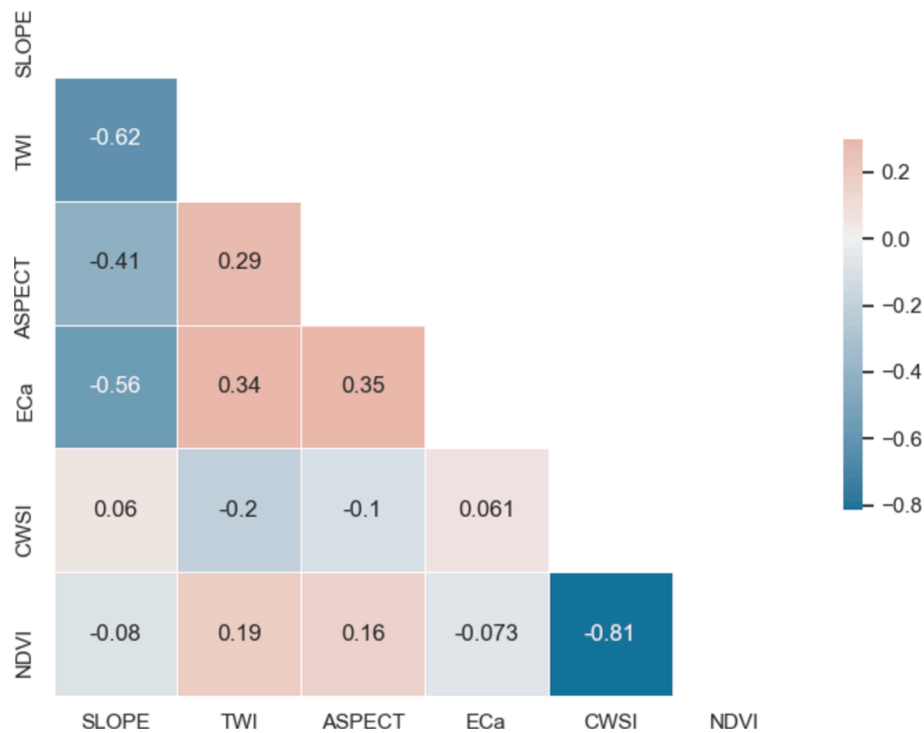


Fig. 4. Correlation matrix (Pearson's correlation coefficients) illustrating the relationships between variables for the data from August 9, 2017. The strongest correlations were found between CWSI and NDVI followed by TWI and Slope and ECa and Slope.

August 15, 2018 and August 29, 2018, with the strongest NDVI-CWSI correlations observed in August for both 2017 and 2018. However, for the data of July 2018, the correlation between NDVI and CWSI was significantly weaker (0.40).

Comparing performance between the different Spatial RF models with the four different geospatial approaches and for the other tested dates (Tables 1-5) suggests the following:

- The best performance for option 1, the Spatial RF model, for 2017 was from August 9 (Table 1) and for 2018 from August 29.
- The lowest performing model for all approaches was found for July 4, 2018 (Tables 1-3) except in approach (c), where July 4, 2018 was the second lowest performing model after August 15, 2018 (Table 4). This can be attributed to the significantly weaker correlations found between CWSI and NDVI, and the fact that the soil at this time had a high moisture content expressed with little variability in values (Ohana-Levi et al., 2022).
- Adding a geospatial component to the model significantly improved its performance and accuracy on all dates. However, while adding (x, y) coordinates as a variable did improve the models, more significant performance improvement was found

when raw data was represented as z-scores, with the approach (b) $G_i^*(x_i)$ resulting in the best performing models followed by approach (c) in which z-scores were used as weights: $(G_i^*(x_i)) \cdot (x_i)$ (Table 5).

- In general, in all the models based on the raw data values, NDVI was the most important variable in predicting CWSI. NDVI was followed by ECa, aspect, or slope in the non-spatial models (Table 1), and by (y/x) in the spatial models, apart from July 2018, (Table 2). The stronger the variable importance of NDVI, the better the model performed, which might also explain why the models of July performed poorly. The weaker the effect of NDVI in predicting CWSI, the stronger the impact of terrain and soil variables. In the models based on z-transformed or weighted values, the most important variable in predicting CWSI was either NDVI or (y/x) (Tables 3 and 4). The high spatial association found between NDVI and CWSI coincides with past research indicating that vegetation indices can be used as proxies for plant water status and for analyzing spatial variability for agricultural management purposes (Bellvert et al., 2014; Helman et al., 2018; Khanal et al., 2017).

Since data scales and units are different in approaches (b) and (c),

Table 1

Performance results for each of the non-spatial models (without any spatial information, used as a reference for evaluating the spatial models). In all the tables 1-5 highlighted values (in dark grey) represent the variables with the highest percentage of importance in predicting the response variable (CWSI), lighter grey values represent the second most important variables in predicting CWSI, R^2 represents how well the model predicts CWSI, MAE is the mean absolute error and RMSE is the root mean squared error.

	VARIABLE IMPORTANCE IN %							ACCURACY MEASURES			
Date and type of model	Slope	Aspect	ECa	NDVI	TWI	Y/X	Response Variable	R ²	MAE	MSE	RMSE
9/8/2017 non-spatial	10.30	10.20	13.49	55.14	10.86	n/a	CWSI	0.685	0.146	0.036	0.189
5/9/2017 non-spatial	15.12	13.56	13.85	44.57	12.90	n/a	CWSI	0.544	0.206	0.070	0.265
4/7/2018 non-spatial	17.95	20.78	19.03	24.60	17.64	n/a	CWSI	0.397	0.188	0.068	0.261
15/8/2018 non-spatial	13.44	12.51	12.94	48.53	12.58	n/a	CWSI	0.574	0.239	0.104	0.322
29/8/2018 non-spatial	11.65	13.58	11.70	53.28	9.79	n/a	CWSI	0.674	0.230	0.085	0.292

Table 2

Performance results for spatial models based on approach (a) in which (x, y) coordinates were added as a variable (y/x).

Date and type of model	VARIABLE IMPORTANCE IN %						Response Variable	ACCURACY MEASURES			
	Slope	Aspect	ECa	NDVI	TWI	Y/X		R ²	MAE	MSE	RMSE
9/8/2017 spatial, approach (a) - (x, y) coordinates	7.66	8.62	9.87	49.00	9.03	15.83	CWSI	0.747	0.146	0.029	0.170
5/9/2017 spatial, approach (a) - (x, y) coordinates	11.21	10.80	10.66	39.72	10.25	17.36	CWSI	0.600	0.192	0.062	0.249
4/7/2018 spatial, approach (a) - (x, y) coordinates	13.77	17.37	14.78	23.63	14.27	16.18	CWSI	0.485	0.171	0.058	0.241
15/8/2018 spatial, approach (a) - (x, y) coordinates	9.56	11.19	9.83	43.26	9.76	16.41	CWSI	0.631	0.224	0.090	0.300
29/8/2018 spatial, approach (a) - (x, y) coordinates	8.35	10.38	9.25	48.45	8.21	15.36	CWSI	0.709	0.218	0.076	0.276

Table 3Performance results for spatial models based on approach (b) $G_i^*(x_i)$.

Date and type of model	VARIABLE IMPORTANCE IN %						Response Variable	ACCURACY MEASURES			
	Slope	Aspect	ECa	NDVI	TWI	Y/X		R ²	MAE	MSE	RMSE
9/8/2017 spatial, approach (b) - $G_i^*(x_i)$	3.99	3.46	6.55	43.59	7.42	34.99	z-transformed CWSI	0.994	0.340	0.207	0.455
5/9/2017 spatial, approach (b) - $G_i^*(x_i)$	4.43	4.48	5.73	32.36	5.96	47.04	z-transformed CWSI	0.994	0.356	0.238	0.488
4/7/2018 spatial, approach (b) - $G_i^*(x_i)$	7.45	25.34	9.66	12.93	14.96	29.67	z-transformed CWSI	0.982	0.370	0.273	0.522
15/8/2018 spatial, approach (b) - $G_i^*(x_i)$	3.38	3.70	6.55	34.92	10.71	40.74	z-transformed CWSI	0.993	0.353	0.241	0.491
29/8/2018 spatial, approach (b) - $G_i^*(x_i)$	3.56	5.02	3.67	40.74	5.38	41.64	z-transformed CWSI	0.996	0.338	0.203	0.450

Table 4Performance results for spatial models based on approach (c) $(G_i^*(x_i)) \cdot (x_i)$.

Date and type of model	VARIABLE IMPORTANCE IN %						Response Variable	ACCURACY MEASURES			
	Slope	Aspect	ECa	NDVI	TWI	Y/X		R ²	MAE	MSE	RMSE
9/8/2017 spatial, approach (c) - $(G_i^*(x_i)) \cdot (x_i)$	6.85	5.96	5.47	41.59	9.33	30.81	Weighted CWSI	0.911	0.912	1.689	1.300
5/9/2017 spatial, approach (c) - $(G_i^*(x_i)) \cdot (x_i)$	6.90	9.35	7.88	30.72	8.09	37.05	Weighted CWSI	0.867	1.306	3.802	1.950
4/7/2018 spatial, approach (c) - $(G_i^*(x_i)) \cdot (x_i)$	11.15	24.58	13.31	13.61	16.19	21.16	Weighted CWSI	0.828	0.632	1.028	1.014
15/8/2018 spatial, approach (c) - $(G_i^*(x_i)) \cdot (x_i)$	7.88	8.23	8.86	32.49	13.25	29.29	Weighted CWSI	0.818	1.509	4.977	2.231
29/8/2018 spatial, approach (c) - $(G_i^*(x_i)) \cdot (x_i)$	6.31	9.16	7.32	40.23	7.28	29.69	Weighted CWSI	0.884	1.698	6.758	2.600

MAE, MSE and RMSE, which were used as accuracy measures in Tables 1–4, are not directly comparable. To compare between the different models a relative error was computed by applying a Min-Max Scaling (Table 5). This enables to evaluate the error in the predictions in relation to the entire data range (eq.5).

$$\text{RelativeError}(\%) = \frac{\text{RMSE}}{(\text{MaxValue} - \text{MinValue})} * 100 \quad (5)$$

Although the best performing models used approach (b), with raw data represented as z-scores, $G_i^*(x_i)$, the models based on approach (c), with z-scores used as weights, $(G_i^*(x_i)) \cdot (x_i)$, were used for testing model predictions of CWSI for a different date. This choice was since, as explained in Eq. (2), predictions based on the $G_i^*(x_i)$ values actually represent the predicted z-transformed (G_i^*) values of CWSI. To present an option that can predict the actual raw CWSI values, for example for management purposes, predictions based on the weighted $(G_i^*(x_i)) \cdot (x_i)$ values were combined in Eq. (2), enabling the output of predicted raw CWSI values at each vine point location.

The model for August 9, 2017 served as the base to predict weighted CWSI values for September 5, 2017 and the model for August 15, 2018

served as the base to predict weighted CWSI values for August 29, 2018. Fig. 5 exhibits the relations between actual and predicted values for September 5, 2017 for each vine.

Calculated correlation between predicted weighted CWSI values and actual (measured) weighted CWSI for each vine was $r = 0.85$.

Predicted values of each individual vine can be also plotted to produce management support maps of CWSI. Since precision irrigation in vineyards is often based on MZs or, as in the present study, on management cells (Fig. 1), averaged predicted CWSI values were used to test prediction accuracy for each cell (Fig. 6). While correlations between actual and predicted weighted CWSI values were significantly higher at the cell level than for the single vines ($r = 0.94$ vs. $r = 0.85$) and in the majority of cells highly correlated, predictions were less accurate for the water-stressed cells B and C. This might be due to model sensitivity to extreme values, as very often the margins of the distribution are less accurately estimated, since there are fewer values for the model to train on. However, the more stressed cells such as K and L resulted in better predictions. An alternative explanation is that cells B and C experienced a relatively higher degree of change in their CWSI values between the two dates, based on measured values, in comparison to the other cells.

Final decisions regarding irrigation should be based on CWSI

Table 5Performance results for models in 2017 comparing the non-spatial (reference) and the spatial models based on approaches (a) to (c). The best performing model was approach (b), based on the R^2 of the validation data (0.994) and on the smallest relative error (1.73).

Date and type of model	VARIABLE IMPORTANCE IN %						Response Variable	ACCURACY MEASURES	
	Slope	Aspect	ECa	NDVI	TWI	Y/X		R ²	Relative Error (%)
9/8/2017 non-spatial	10.30	10.20	13.49	55.14	10.86		CWSI	0.685	11.4
9/8/2017 spatial, approach (a) - (x, y) coordinates	7.66	8.62	9.87	49.00	9.03	15.83	CWSI	0.747	10.26
9/8/2017 spatial, approach (b) - z-scores	3.99	3.46	6.55	43.59	7.42	34.99	z-transformed CWSI	0.994	1.73
9/8/2017 spatial approach (c) - z-scores as weights	6.85	5.96	5.47	41.59	9.33	30.81	Weighted CWSI	0.911	4.56

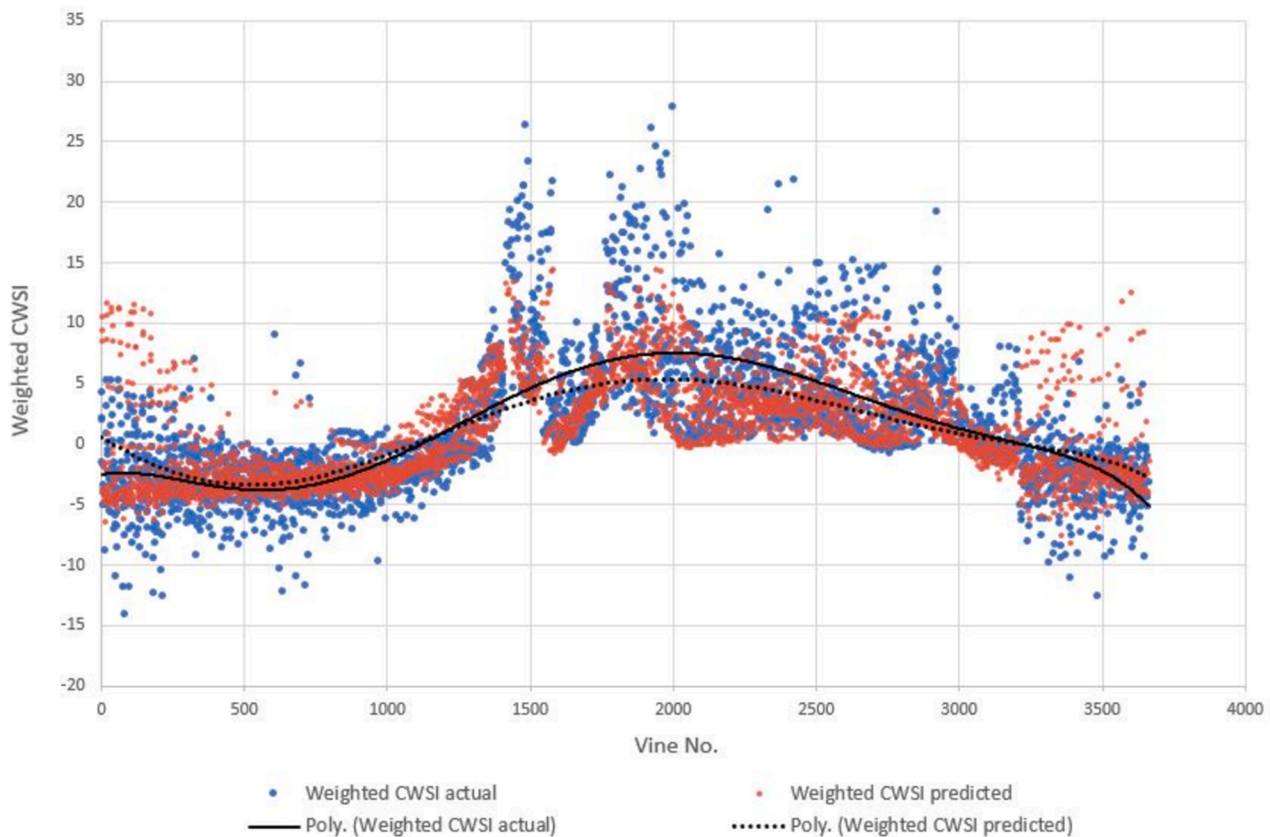


Fig. 5. Differences between predicted vs. actual (measured) weighted CWSI values, for each vine, based on model for August 9, 2017, which was used to predict weighted CWSI values for September 5, 2017) according to approach (c) $(G_i^*(x_i)) \cdot (x_i)$.

thresholds for each phenological stage (Prueger et al., 2019). Irrigation decisions based on CWSI thresholds have been quantified and used in various crops such as grapevines (Bahat et al., 2021), peaches (Katz et al., 2023), and apples (Osroosh et al., 2016). Current research characterizing CWSI for precision irrigation in vineyards has shown substantial diurnal fluctuations, although trends remained similar (Prueger et al., 2019). Thus, more research is required in order to quantify these diurnal and seasonal fluctuations for specific case studies with specific environmental conditions and accordingly define the required thresholds along the growing season. While the current research suggests optimizing the use of remote sensing to support precision irrigation decisions, use of NDVI to correlate with CWSI may not always be successful or may benefit in combination with additional accessible plant and/or water related indices (Girolimetto & Venturini, 2013; Helman et al., 2018). The relationship between SWP and CWSI is stronger and better documented, but has sometimes been found inconsistent throughout the season in Mediterranean vineyards (Gonzalez-Dugo et al., 2013; Möller et al., 2007). Therefore, using NDVI to predict CWSI and associating CWSI as a proxy for SWP thresholds for irrigation decision-making require further research.

4. Summary and conclusions

The current research presented a methodology for generating in-season maps of predicted CWSI in a vineyard. This was accomplished by combining static and spatial-dynamic variables through the implementation of random-forest machine learning prediction. A primary motivation of the study was to explore the potential for minimizing expenses associated with thermal-imaging campaigns. This was attempted by supplementing thermal image-generated CWSI maps with more cost-effective multi-spectral-generated NDVI maps. To enhance

CWSI prediction accuracy, the spatial dimension was introduced to an RF model. A spatial machine-learning model was developed to quantify the spatial variability of water status and support irrigation decisions for wine-grape vineyards by predicting CWSI values for single vines. Due to their flexibility and ability to accurately simulate complex non-linear relations between variables, ML models can reliably benefit decision-making in commercial agriculture. When coupled with spatial-statistical methods, spatial structure can be modelled, and complex spatial processes, such as autocorrelation and heterogeneity, can be accounted for. The model was applied to a heterogenous commercial vineyard in Israel. Our results and performance evaluation indicated that such models are viable for predicting CWSI prior to harvesting. The best performing models were those that predicted the spatial pattern of CWSI, i.e., those based on the z-transformed values, indicating that accounting for the geospatial structure in an RF model enhances the accuracy of the model as well as its predictions. It was found that it is not sufficient to introduce (x, y) coordinates as predictive variables. Instead, to fully account for spatial autocorrelation, spatial relationships must be represented directly within the models, such as by using raw data transformed into z-scores. In addition, CWSI prediction models based on a limited number of variables can perform sufficiently well in terms of their accuracy. They can be used as a basis for developing practical user-friendly DSTs, to be applied by farmers and decision-makers to produce irrigation prescription maps based on readily available, cost-efficient remotely-sensed NDVI data with high temporal and spatial resolution. Such models can reduce not only the number of thermal-imaging campaigns per season, but also the number of SWP in-situ measurements required during the growing season, consequently reducing management costs substantially. Locations and dates where predictions are less accurate are also important indicators, as they can point to decisions regarding sensor/sampling distribution and be used to optimize

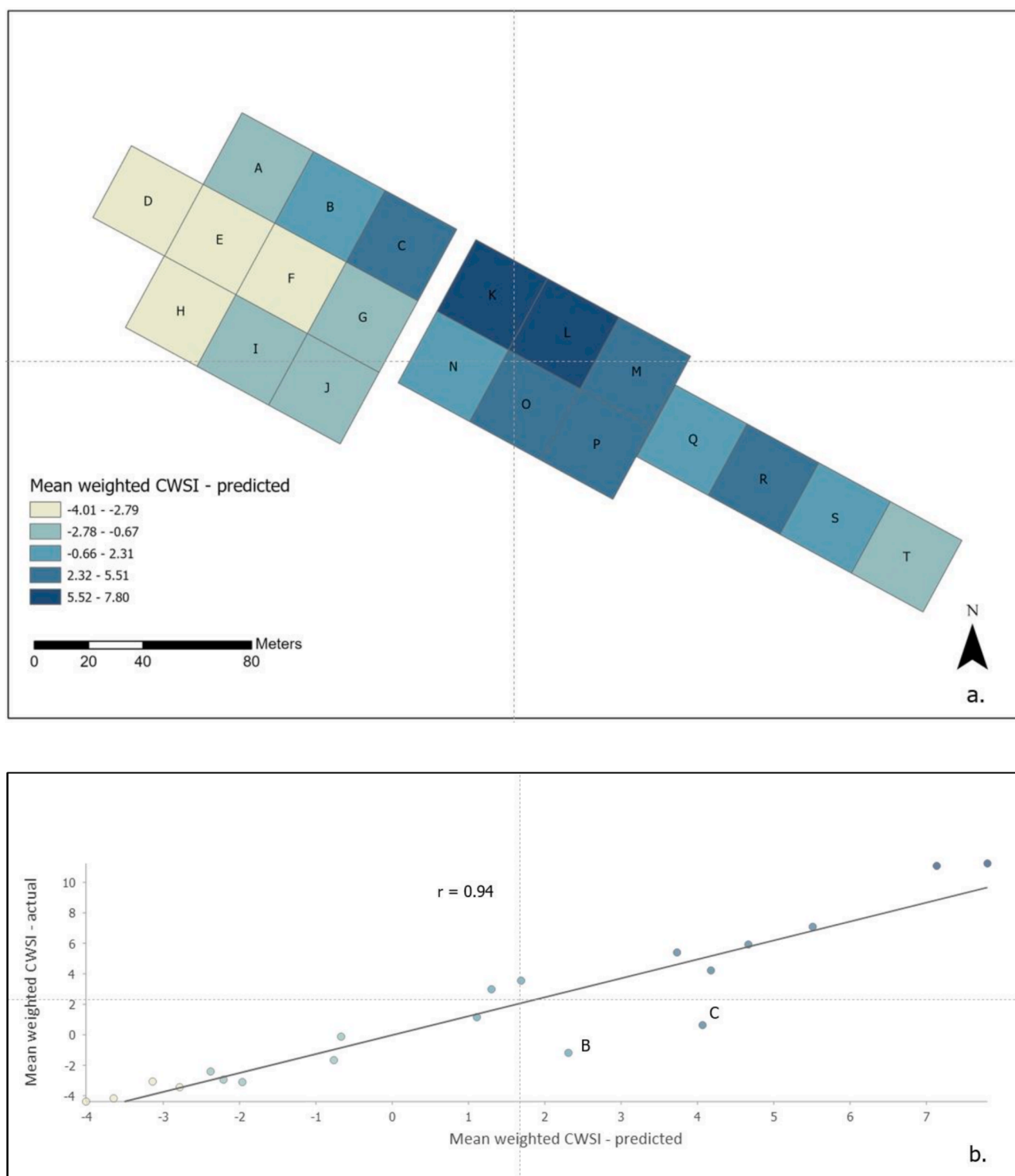


Fig. 6. Spatial distribution of predicted weighted CWSI values for September 5, 2017 based on the model of August 9, 2017 averaged per management cell (a), and correlation between predicted and measured (actual) weighted CWSI values averaged per management cell (b), indicating cells B and C where predictions were less accurate.

decisions regarding the dates of thermal-imaging campaigns.

Input variables used for developing the model can be substituted depending on the specific conditions and management requirements of the vineyard. For example, previous seasons' yield maps, measurements of plant vine size, if monitored with sufficient spatial distribution, or any other relevant spatial data, may be considered as predicting variables for water requirements. Substituting the variables will, of course, necessitate additional research and adjustment of model parameters in order to

arrive at the best performing model. To reduce the probability of over-fitting to specific input data and provide a general-purpose model that can perform well in other vineyards, future work must account for geographical and environmental variability (e.g., grapevine variety, climatic region, soil type, average chilling hours, etc.). The same, of course, is true regarding evaluation of the fitness of the method for additional crops including fruit tree orchards or vegetable or field crops for which plant water status is used in irrigation BMPs. Socio-economic

and cost-benefit considerations should also be integrated to examine whether precision irrigation management based on such models has the desired effects on water consumption and yield compared to standard practices.

To conclude, a spatial machine-learning model based on an RF algorithm combined with spatial statistical methods was developed to predict the spatial variability of CWSI in a vineyard. The predicted CWSI based on relatively easily accessible maps of DEM-derived slope, aspect and TWI, soil ECa, and NDVI was found reliable. Inclusion of a geospatial component significantly improved model performance and accuracy, particularly when raw data was represented as z-scores or when z-scores were used as weights. In the case for an experimental vineyard in Israel, CWSI prediction by the spatial model was most strongly influenced by the input data NDVI, followed by individual vine location. Model performance was enhanced as a function of increasing relative importance of the temporally changing variable NDVI on CWSI. When the effect of NDVI in predicting CWSI was weaker, terrain and soil variables had more substantial influence. The model, and future versions of it based on alternative predicting and predicted variables should be beneficial to irrigation scheduling and as a basis for VRI prescription maps.

CRediT authorship contribution statement

Aviva Peeters: Conceptualization, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Yafit Cohen:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Writing – original draft, Writing – review & editing. **Idan Bahat:** Conceptualization, Data curation, Methodology, Project administration, Resources, Writing – original draft, Writing – review & editing. **Noa Ohana-Levi:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Eitan Goldstein:** Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Yishai Netzer:** Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Writing – original draft, Writing – review & editing. **Tomás Roquette Tenreiro:** Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Victor Alchanatis:** Funding acquisition, Project administration, Resources, Writing – review & editing. **Alon Ben-Gal:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2024.109578>.

Data availability

Data will be made available on request.

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